Saqya ساقية

The right amount of water, on your canal's schedule, in a text message.

Eyad Selim · solo · final-year secondary school student · Cairo, Egypt

Live prototype: **eyad46.github.io/saqya**

THE PROBLEM

Egypt needs far more water than it has, and farming takes most of it

65 vs 88.5

billion m³ a year: renewable water against total demand

≈500 m³

per person per year, less than half the 1,000 m³ line used to define water scarcity

79%

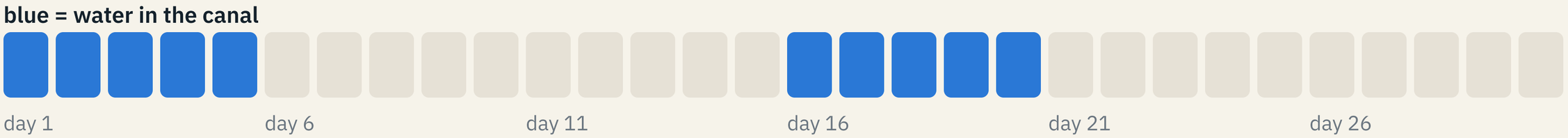
of all the freshwater Egypt withdraws goes to agriculture

Every cubic metre a farm does not waste is water for someone else.

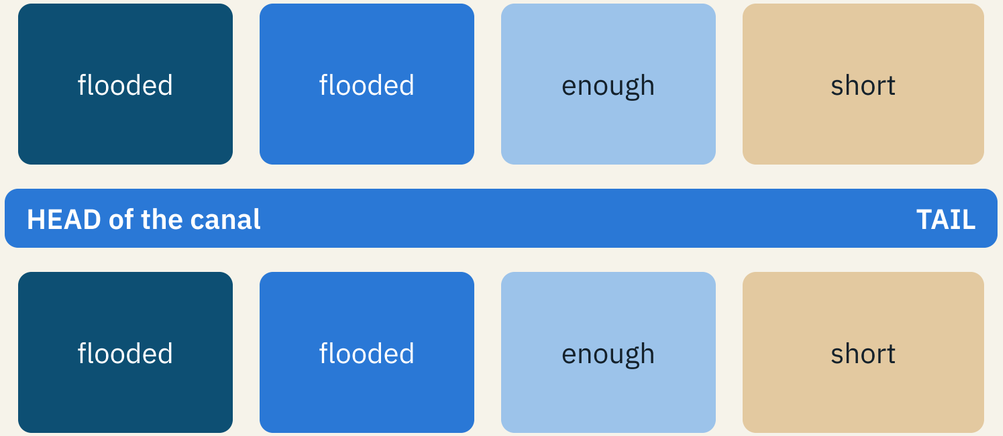
Sources: Minister of Water Resources and Irrigation Hani Sewilam, Ahram Online, 7 April 2026; Ahram Online, "Egypt's per capita water drops to under half of UN threshold", 2026; World Bank indicator ER.H2O.FWAG.ZS (FAO AQUASTAT), 2022.

WHY WATER IS WASTED

In the Delta, a field only gets water on its canal's turn



- Winter rotation: about **5 days on, 10 days off**.
- Unsure when water will come again, farmers at the head of the canal **over-irrigate**, and farmers at the tail get too little.
- Conventional wheat: **5 floods of about 435 m³ per feddan**, whatever the weather.



Sources: FAO, Rapid assessment study, irrigation and drainage in Egypt (IIIMP): "5 days on/10 days off in winter"; IWMI (2015) Heads or Tails, Research Report 160; El-Hadidi et al. (2016), Sakha field trial, conventional 5 irrigations: 2,173 m³ per feddan.

WHAT EXISTS TODAY

Existing tools don't fit a farmer on a rotation canal

	SOIL SENSORS	SMARTPHONE APPS	SAQYA
Hardware in the field	**yes, per field**	**no**	**no**
Needs a smartphone and mobile data	often	**yes**	**no, plain SMS on any phone**
Plans around the canal's turns	**no**	**no, they give an interval between irrigations**	**yes, decides at every turn**
Tells the farmer what to do	a moisture reading	irrigation hours	**skip, or pump for exactly N minutes**

IWMI and FAO's IRWI app (2020) does the same water science for Delta farmers, but its developers said weak internet in the fields was their biggest obstacle.

Source: "Innovative phone app helps Egypt farmers enhance water, soil productivity", Zawya, 2020; IWMI brief "Irrigation scheduling advice: Egypt", 2025.

THE SOLUTION

One text message before every canal turn

Skip this turn. The soil holds enough water for your wheat until your next turn.

Light irrigation. Run your pump 1 h 25 min only. Don't flood the field.

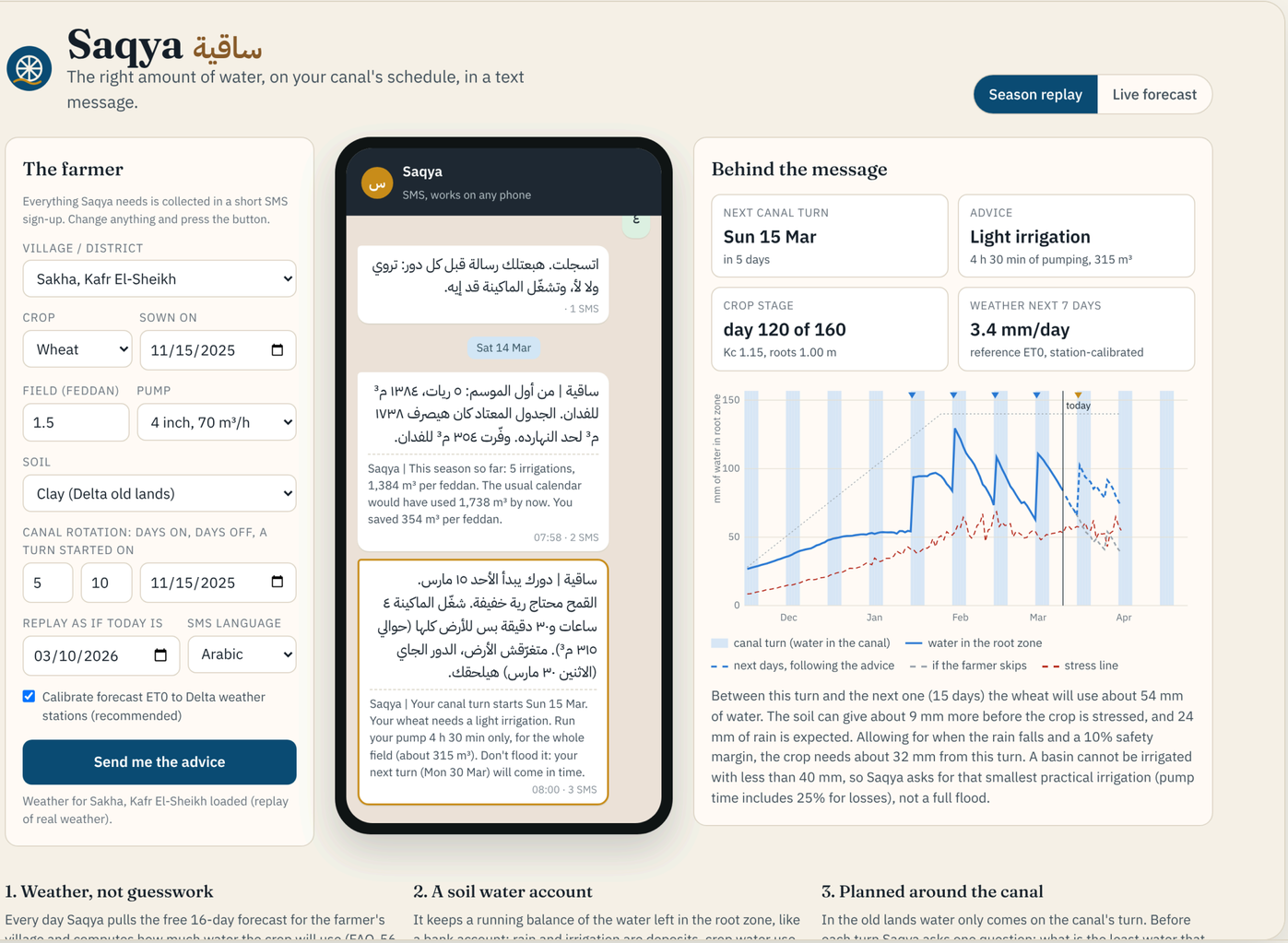
Full irrigation. Hot days ahead and your next turn is far.

any phone | Egyptian Arabic | no app, no internet, no sensors



WORKING PROTOTYPE

Live forecasts, real advice, today



1. Weather, not guesswork

Every day Saqya pulls the free 16-day forecast for the farmer's village and computes how much water the crop will use (FAO-56

2. A soil water account

It keeps a running balance of the water left in the root zone, like a bank account: rain and irrigation are deposits, crop water use

3. Planned around the canal

In the old lands water only comes on the canal's turn. Before each turn Saqya asks one question: what is the least water that

Try it: **eyad46.github.io/saqya**

Any village, crop, canal schedule and pump. Live forecast, or a replay of any past season.

HOW SAQYA DECIDES

Standard science, one new question

1

Weather. Free 16-day forecast for the farmer's village (Open-Meteo): reference evapotranspiration and rain, calibrated to Delta weather stations.

→

2

Water account. FAO-56 crop coefficients turn weather into crop water use by growth stage. Rain and irrigation in, crop use out, and the soil only holds so much.

→

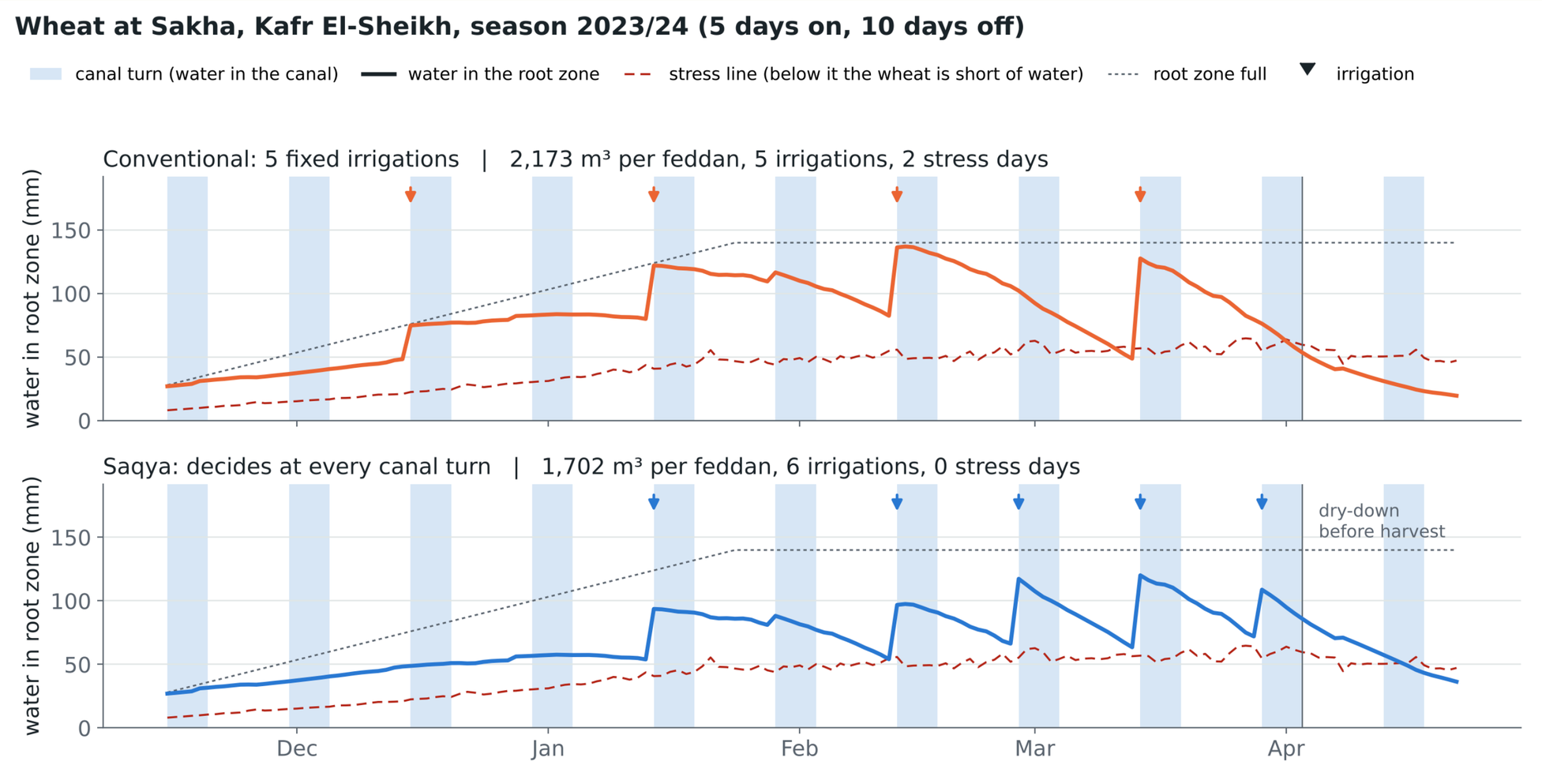
3

Canal-aware plan. At each turn: the least water that keeps the crop above its stress line until the next turn. Sent as pump minutes for the farmer's field.

Method: Allen et al. (1998), FAO Irrigation and Drainage Paper 56, Tables 11, 12, 19 and 22. ET0 from reanalysis is 10 to 25% above Delta station values, so it is scaled month by month to station data.

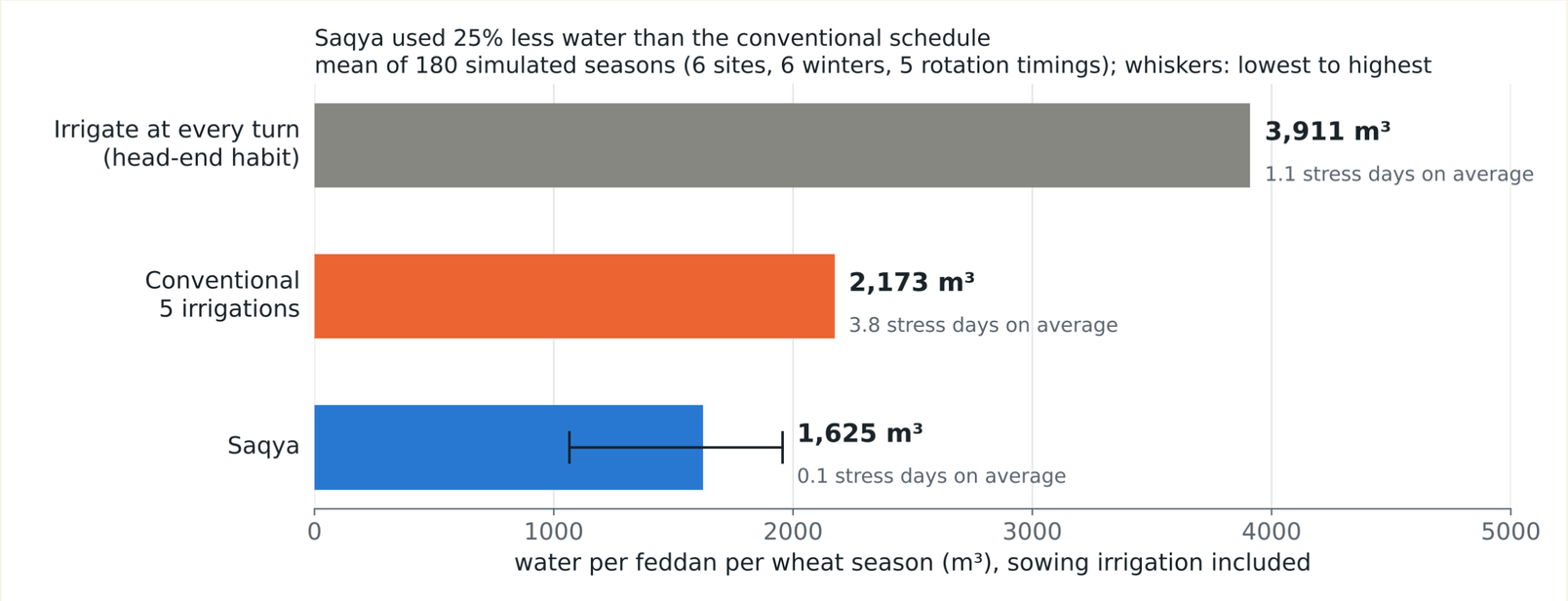
THE NEW PART

Same field, same weather, same canal



RESULTS

Tested on 180 real seasons: 25% less water



548 m³
saved per feddan per wheat season

180 / 180
seasons where Saqya used less water

0.1 vs 3.8
stress days, Saqya vs conventional

Real weather 2019 to 2025 at 6 Delta sites, 5 rotation timings. Saqya decided with a 7-day forecast with errors added and long-term averages after that. Conventional: 5 irrigations of 435 m³ per feddan (Sakha trial). Both at 80% application efficiency. If crop demand were 20% higher, the saving would be 6%.

IMPACT

What it could mean for Egypt

3.7 M

feddans of wheat in the 2025/26 season

55 M m³

saved every winter if Saqya reached 100,000 of them: the yearly water share of about 110,000 Egyptians

≈8 h

less pumping per feddan per season with a 4-inch pump, so less diesel

Water left in the canal reaches farmers at the tail. Part of the water a farm saves would otherwise return to drains and be reused, so savings for the whole Nile system are smaller than on the farm.

Sources: Ministry of Agriculture via Egyptian Cabinet (MEObserver, 14 April 2026); 500 m³ per person (Ahram Online, 2026); 548 m³ per feddan from the simulation; pump of 70 m³ per hour.

FEASIBILITY

Cheap to run, ready to pilot

About $1 to $2 per farmer per season. 12 to 15 messages, 2 to 3 SMS parts each in Arabic, at $0.025 to $0.04 per SMS.

About $0.002 per cubic metre saved.

Free weather data, a public method, open-source code. No hardware to buy or repair.

1. Pilot (one season): 30 farmers on one canal in Kafr El-Sheikh with a water users' association; soil moisture checks on 10 fields to calibrate.

2. Grow: 1,000 farmers through extension workers; WhatsApp voice notes for farmers who do not read.

3. Scale: irrigation district offices send it to whole canals; more crops.

SMS prices to Egypt from a provider comparison (sent.dm, prices checked January 2025). Cost per m³ assumes 1.5 feddan and 548 m³ saved per feddan.

LIMITS AND NEXT STEPS

Saving water can start with one text message

- The model does not yet include the shallow water table or salinity.
- Soil water holding and the smallest practical irrigation need calibration in real fields.
- The results are a simulation with real weather; the next step is a field pilot.

Thank you. شكراً

Live demo: **eyad46.github.io/saqya** · Code: **github.com/Eyad46/Eyad46.github.io/tree/main/saqya**

Eyad Selim, solo, Cairo. Outside help: AI assistants were used to help write code and edit text. All data sources and methods are cited.